

OLIVER SHAPIRO

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PROFESSIONAL SUMMARY

Mechanical engineer with hands-on experience in prototyping, manufacturing, simulation, and electromechanical integration. Comfortable taking physical products from concept to commercial launch on small, fast-moving teams.

EDUCATION

Dartmouth College, Hanover, NH

June 2026

Bachelor of Engineering (Mechanical), Minor in Applied Mathematics

GPA 3.8

Honors/Awards: 2024-2025 Rufus Choate Scholar (top 5% of all students for academic year)

PROFESSIONAL EXPERIENCE

Hextronics, Inc., Miami, FL

June 2026—September 2026

20-person, Series-A engineering startup developing autonomous drone-in-a-box infrastructure

Mechanical Engineering Intern

- Led fabrication of drone docking station prototypes for product debut with industrial 3D printers
- Designed FAA-compliant lighting system for industrial inspection drone that meets 3-mile visibility requirement
- Developed and manufactured precision assembly jig for autonomous drone charger, reducing installation time by 70%
- Coordinated with electrical engineering team to design and implement custom printed circuit boards
- Applied DFM principles and worked with suppliers to optimize production parts for injection molding and CNC machining
- Designed weather station integrating piezoelectric and Hall-effect sensors that streams data to flight management system

Thayer School Machine Shop, Dartmouth College, Hanover, NH

June 2023—June 2026

Senior Teaching Assistant

- Fabricated parts for researchers in quantum mechanics, biomedical imaging device labs, and others
- Instructed students and faculty on shop safety and fabrication skills, including CNC mill/lathe, welding, 3D printing, CAD

Rainier Mountaineering, Inc., Ashford, WA

May 2025—August 2025

Mountain Guide

- Led clients to the summit of Mount Rainier, the most prominent and glaciated peak in the contiguous United States
- Maintained the primary summit route via crevasse ladder installation and hazard-driven rerouting

Professional Ambulance & Oxygen Service, Cambridge, MA

January 2022—August 2022

Emergency Medical Technician

- Provided patient care and drove ambulances in 911 emergency situations

SELECTED PROJECTS

Racing Drone and Flying Wing Assembly

Personal Project

- Assembled electromechanical propulsion systems (ESCs, flight controllers) for six FPV quadcopter and flying wing builds
- Optimized flight behavior through PID tuning and motor/propeller performance testing

Cold-Weather Camping Stove

September 2025—March 2026

Undergraduate Capstone Project

- Investigated cold-weather failure modes of camp stoves and designed electromechanical pressure regulator to mitigate them
- Applied thermal FEA to model heat transfer across temperature ranges from -40°C to ambient

Fracture Energy Analysis of 3D Printed Polylactic Acid (PLA) Polymers

March 2024—May 2024

Undergraduate Research Project

- Employed ASTM standard Charpy impact testing methods to investigate the ductile-brittle transition of PLA polymers
- Utilized Scanning Electron Microscopy to characterize PLA failure modes at different temperatures

Kangaroo Belt Emergency Seatbelt System

March 2024—June 2024

Undergraduate Design Project

- Designed auto-locking, auto-retracting seatbelt that lowers patient stretcher securing time by 66% in emergencies
- Performed FEA analysis and physical tensile tests to validate adherence to national Title 49 strength requirements

LEADERSHIP & OTHER ACTIVITIES

Dartmouth Lightweight Rowing Team

September 2023—June 2026

Varsity Division I Athlete, Assistant Captain

- Trained 15–20 hours/week; selected as assistant captain and liaison to coaching staff to enhance team cohesion

Arava Institute for Environmental Studies, Ketura, Israel

September 2021—January 2022

- Residential program in environmental and regional peace studies; researched the ecological makeup of Israel's Arava Valley

SKILLS & CERTIFICATIONS

CAD: SOLIDWORKS (CSWA certified, experienced with CAM/HSMWorks, drawings, GD&T), Fusion 360

Analysis & Simulation: SOLIDWORKS Flow CFD, SOLIDWORKS FEA, Fusion Simulation

Programming: MATLAB, Python, C, Java

Technical Skills: CNC lathe/mill, CAM toolpath generation, FDM/SLA printer repair/operation, TIG welding, Instron tensile testing

Languages: Spanish

INTERESTS

Crossword puzzles, rock climbing, mountaineering, trail running, backcountry skiing